

Jeonghyeon Kim

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RESEARCH INTEREST

My primary research goal is to build reliable and trustworthy AI systems without compromising their predictive performance. To achieve this, I actively conduct research on Out-of-Distribution (OoD) detection, machine unlearning, multi-modal fine-tuning, and concept erasure. I particularly enjoy interpreting these problems through the lens of Energy-Based Models (EBMs) to provide theoretical grounding for empirical results.

EDUCATION

Seoul National University of Science and Technology (SeoulTech)	Seoul, South Korea
<i>Ph.D. Student in Data Science (Advisor: Prof. Sangheum Hwang)</i>	<i>Mar. 2023 - present</i>
Seoul National University of Science and Technology (SeoulTech)	Seoul, South Korea
<i>B.S. in Industrial Engineering</i>	<i>Mar. 2016 - Feb. 2022</i>

RESEARCH EXPERIENCE

DAINTLAB, SeoulTech	Seoul, South Korea
<i>Research Assistant (Advisor: Prof. Sangheum Hwang)</i>	<i>Aug. 2022 - present</i>

- **Generative AI Safety & Concept Erasure:**
 - Addressed the trade-off between removing harmful concepts and preserving generation quality in Diffusion Models. Proposed **High-level Representation Misdirection (HiRM)**, a method for localized concept erasure by misdirecting semantic representations in the text encoder.
 - Developed a strategy to update only early layers while guiding high-level semantics, achieving precise removal of target concepts (e.g., styles, objects, NSFW) with minimal impact on unrelated concepts.
 - ***Selected Publication: Accepted at ICLR 2026***
- **Trustworthy Multimodal Representation Learning:**
 - Proposed **Cross-Modal Alignment (CMA)** to enhance Out-of-Distribution (OoD) detection in Vision-Language Models (VLMs) like CLIP.
 - Theoretically demonstrated that the CMA regularization corresponds to maximizing the likelihood of a joint Energy-Based Model (EBM) on a hyperspherical space.
 - Achieved state-of-the-art OoD detection performance by mitigating the modality gap within ID embeddings while maintaining high ID accuracy.
 - ***Selected Publication: Accepted at CVPR 2025***

Machine Learning Data Research Group, KISTI	Daejeon, South Korea
<i>Research Intern (Advisor: Dr. Ryong Lee)</i>	<i>Jul. 2021 - Aug. 2021</i>

- **Privacy-Preserving Pipeline for Autonomous Driving:**
 - Built a data de-identification pipeline using object detection (YOLOv3) to protect personal information (faces, license plates) in large-scale driving datasets.

SKILLS

Languages & Frameworks Linux, Python, PyTorch, Hugging Face, Docker
Research Areas Concept Erasure, Machine Unlearning, OoD Detection, Multimodal Learning

PUBLICATIONS

International Conferences

[IC4] “*Localized Concept Erasure in Text-to-Image Diffusion Models via High-Level Representation Misdirection*”

Uichan Lee*, **Jeonghyeon Kim*** and Sangheum Hwang

International Conference on Learning Representations (**ICLR**), 2026

[IC3] “*Enhanced OoD Detection through Cross-Modal Alignment of Multi-Modal Representations*”

Jeonghyeon Kim and Sangheum Hwang

IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2025

[IC2] “*Comparison of Out-of-Distribution detection Performance of CLIP-based Fine-Tuning Methods*”

Jeonghyeon Kim, Jihyo Kim, and Sangheum Hwang

International Conference on Electronics, Information, and Communication (**ICEIC**), 2024 (**Oral, Best Paper Award**)

[IC1] “*Deep Active Learning with Contrastive Learning Under Realistic Data Pool Assumptions*”

Jihyo Kim, **Jeonghyeon Kim**, and Sangheum Hwang

AAAI Workshop on Practical Deep Learning in the Wild (**AAAI-W**), 2023

Domestic Conferences

[DC5] “*Reference-free LLM Unlearning Evaluation through Energy Trajectory Analysis*”

Jeonghyeon Kim, and Sangheum Hwang

Korea Computer Congress (**KCC**), 2026 (**Oral, Best Paper Award**)

[DC4] “*Parameter-Efficient LLM Unlearning through Sparse Attention Head Updates*”

Deokyong Lee, **Jeonghyeon Kim**, and Sangheum Hwang

Korean Institute of Industrial Engineers (**KIIE**), 2025

[DC3] “*Uncovering Hidden Vulnerabilities in Machine Unlearning: Adversarial Attack as a Probe and Pruning as a Solution*”

Hwiyeong Lee, **Jeonghyeon Kim**, and Sangheum Hwang

Korea Computer Congress (**KCC**), 2024

[DC2] “*Uniformity Reduction Regularization: Multimodal Fine-tuning to Enhance Out-of-Distribution Detection Performance*”

Jeonghyeon Kim and Sangheum Hwang

Korea Computer Congress (**KCC**), 2024 (**Oral**)

[DC1] “*Enhancing Out-of-Distribution Detection Performance of CLIP Based on Fine-tuning with Random Texts*”

Jeonghyeon Kim, Jihyo Kim, and Sangheum Hwang

Korea Computer Congress (**KCC**), 2023 (**Oral, Best Paper Award**)

INVITED TALKS

KCC 2026	Localized Concept Erasure in Text-to-Image Diffusion Models via High-Level Representation Misdirection (Oral, presenter : Uichan Lee)	<i>Jun. 2026</i>
KCC 2026	Enhanced OoD Detection through Cross-Modal Alignment (Oral) (Jeju, ICC Jeju)	<i>Jun. 2026</i>
KCCV 2025	Enhanced OoD Detection through Cross-Modal Alignment (Poster) (Busan, Bexco)	<i>Jul. 2025</i>
KSIIM 2025	Enhanced OoD Detection through Cross-Modal Alignment (Oral) (Seoul, Seoul National University College of Medicine)	<i>Feb. 2025</i>

HONORS AND AWARDS

Best Paper Award	KCC 2026	<i>Jun. 2026</i>
Ph.D. Student Research Grant	National Research Foundation of Korea (NRF)	<i>Aug. 2025</i>
- Grant Amount: 50M KRW / 2 years		
- Topic: Mechanistic Interpretability of Knowledge Unlearning in Foundation Models		
Best Paper Award	ICEIC 2024	<i>Jan. 2024</i>
Best Paper Award	KCC 2023	<i>Jun. 2023</i>
Gold Prize	Capstone Competition, Dept. of IISE, SeoulTech	<i>Dec. 2021</i>
School Honor Scholarship	SeoulTech	<i>Fall 2021</i>
Academic Excellence Scholarship	SeoulTech (4 semesters)	<i>2016 - 2021</i>

LICENSES AND CERTIFICATIONS

Engineer Quality Management	Human Resources Development Service of Korea	<i>Dec. 2020</i>
Engineer Information Processing	Human Resources Development Service of Korea	<i>May. 2019</i>

TEACHING EXPERIENCE

Neural Networks and Deep Learning	SeoulTech
<i>Teaching Assistant (Instructor: Prof. Sangheum Hwang)</i>	<i>Spring 2026</i>
Deep Learning	SeoulTech
<i>Teaching Assistant (Instructor: Prof. Sangheum Hwang)</i>	<i>Spring 2024</i>
Artificial Intelligence	SeoulTech
<i>Teaching Assistant (Instructor: Prof. Sangheum Hwang)</i>	<i>Fall 2023</i>

LANGUAGES

Korean : Native, **English** : Proficient